

Sharad Kumar Tiwari

Machine Learning Engineer

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Professional Summary

AI/ML Engineer with 5 years of experience delivering production-grade ML, GenAI, NLP, and MLOps systems at scale in product environments. Built LLM-powered solutions processing 2,000+ daily customer interactions, reducing operational effort by up to 80% and enabling faster complaint resolution to support customer retention. Experienced across the full ML lifecycle and cloud infrastructure on AWS, Azure, and Snowflake.

Professional Experience

Jet2 Travel Technologies Pvt. Ltd. (Jet2 PLC), Pune, Maharashtra

Machine Learning Engineer (July 2024 – Present)

1. Designed a production GenAI intent classification system with Snowflake Cortex AI and fine-tuned Mixtral 8x7B, processing 2,000+ customer call transcripts/day, reducing manual categorization effort by 80%, improving call routing accuracy, and directly enhancing customer resolution experience.
2. Built a real-time multi-label complaint triage system on AWS Bedrock (18 classes) with auto-summarization and a serverless event-driven pipeline using AWS Lambda, SQS, and SNS, reducing manual triaging effort by ~70% and enabling faster complaint resolution, supporting customer retention outcomes.
3. Developed an internal AI-powered HR sourcing and screening system using CrewAI and AWS Bedrock, featuring specialized agents for JD analysis, resume parsing, candidate-JD gap analysis, and automated interview evaluation - deployed on AWS ECS via Docker, ECR & CloudFront; currently being presented to CTO for organization-wide adoption.
4. Developed a document processing pipeline using FastAPI, OCR, and PDF parsing to automate work-order extraction, reducing processing time from 5–10 hours to under 5 minutes — directly unblocking revenue-critical operational workflows.
5. Architected a reusable Amazon SageMaker MLOps framework for near-AutoML training and inference - covering automated retraining, cross-account model sharing, blue-green deployments, Docker-based container frameworks, Azure DevOps CI/CD pipelines, and an internal Python utility library integrating Snowflake, BigQuery, GCS, and AWS services - reducing ML workflow onboarding time by 50%, enabling production delivery for 10+ data science projects, and accelerating DS experimentation cycles by ~40%.

Codiant – A Yash Technologies Company, Indore, Madhya Pradesh

Machine Learning Engineer (Jan 2022–July 2024)

1. Launched a queue-based RAG Q&A pipeline using Azure OpenAI GPT-3.5, Ada embeddings, LangChain, FAISS, and prompt engineering (zero-shot, few-shot) - reducing enterprise document lookup from 5+ days to minutes and accelerating business decision and contract turnaround cycles.
2. Developed an end-to-end document detail extraction system using a custom YOLO + OCR + image classification pipeline, achieving 80%+ accuracy; enabled stakeholder validation via a Streamlit app.
3. Automated document classification and tagging pipeline using SpaCy NER, Elasticsearch, and a classification ML model, achieving 93% field-level accuracy (detail extraction F1: 0.89, document tagging F1: 0.93).
4. Engineered a resume intelligence engine using SpaCy NER/REL deployed on AWS Lambda, auto-processing 1,000+ resumes/month and eliminating manual screening effort for recruiters.
5. Built a multimodal recommendation system leveraging pre-trained transformer models to extract embeddings from text, images, video, and audio, using a vector database for scalable similarity retrieval.

Pojo Software Pvt Ltd, Chennai, Tamil Nadu

Data Analyst Intern (Sep 2021 – Jan 2022)

Developed and back-tested high-frequency trading algorithms using statistical analysis and mathematical modeling to optimize performance and reduce portfolio volatility.

Personal Projects

AppointEase - Clinic Appointment ChatBot & Analytics Platform | RASA, Python, Streamlit, MongoDB, Nginx

1. Built a Rasa-based conversational AI system for clinic appointment management supporting create, modify, cancel, and emergency booking intents; handles patient intake including name, DOB, contact, gender, symptoms, existing conditions, and current medications. Implemented an intelligent slot scheduling engine based on real-time doctor availability
2. Developed a clinic analytics dashboard with daily, weekly, and monthly appointment metrics, operational KPIs, and a patient record lookup interface for doctors; deployed on local infrastructure using Nginx.
3. Rasa ChatBot Repo - <https://github.com/sharadktiware/patient-appointment-rasa-chatbot>, Streamlit Dashboard Repo - https://github.com/sharadktiware/doctor_clinic_dashboard_streamlit .

Technical Skills

Generative AI & LLMs: LLMs, RAG, Prompt Engineering, LangChain, LangGraph, LlamaIndex, CrewAI, Hugging Face, Fine-Tuning, Embeddings, Agentic AI, RASA, Ollama, AWS Bedrock, Snowflake Cortex AI, Azure OpenAI

ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, CatBoost, LightGBM, Transformers, Recommendation Systems, Time Series Forecasting, Statistical Modeling, Model Explainability (Shap), Feature Engineering, Predictive Modeling

NLP & Vision: spaCy, NER, Word Embeddings, OCR, Object Detection, YOLO

MLOps & Cloud: AWS, Azure, GCS, Amazon SageMaker, Azure ML Studio, Azure DevOps, Docker, MLflow, Terraform, DVC, Kubeflow, CI/CD, Model Monitoring, Model Deployment

Vector Databases: FAISS, Chroma, Elasticsearch, Pinecone

Data & Infrastructure: Snowflake, BigQuery, PostgreSQL, MySQL, MongoDB, Redis, SQLite, AWS S3

Languages & Tools: Python, SQL, FastAPI, Streamlit, Flask, Django, Git, Jupyter, VS Code, Tableau, Dataiku DSS, Postman

Certifications

1. **AWS Associate Machine Learning** - [AWS Skill Builder](#) (Sep-2025)
2. **AWS Technical Essentials** - [AWS Skill Builder](#) (Dec-2025)
3. **Deep Learning Specialisation (Neural Networks & Hyperparameter Tuning)** - [deeplearning.ai](#) (Jul-2023)
4. **Machine Learning Specialisation** - [deeplearning.ai](#) (May-2023)
5. **DP-100: Data Science on Microsoft Azure Exam** - [coursera.org](#) (Jan-2023)
6. **Natural Language Processing in Microsoft Azure** - [coursera.org](#) (Dec-2022)

Education & Certifications

Bachelor of Engineering in Computer Science & Engineering - Jiwaji University, Gwalior
8.2 / 10 CGPA | 2017 – 2021

Achievements

Employee of the Quarter (Jan - Mar 2026) - Jet2 Travel Technologies | 2026

Diamond Spot Award - Jet2 Travel Technologies | 2025

Gold Medal for Academic Excellence - Jiwaji University | 2022